CBSE 2027 Chemistry — Predicted Paper (Set 9 of 10)

Based on the official CBSE blueprint, extensive PYQ analysis, and high-frequency topics from the NCERT syllabus for 2026-27, here is Set 9 of the predicted board paper. All questions are entirely fresh, rigorously tested concepts that have a high probability of appearing in the upcoming board exam.

SECTION A: Multiple Choice & Assertion-Reason Questions (1 Mark Each x 16 = 16 Marks)

Q1. A mixture of nitric acid (HNO_3) and water forms a maximum boiling azeotrope. This indicates that the mixture: (a) obeys Raoult's law perfectly. (b) shows a positive deviation from Raoult's law. (c) shows a negative deviation from Raoult's law. (d) behaves as an ideal solution at all concentrations. * **Chapter:** Solutions | **Confidence:** Very High

Q2. The relationship between conductivity (κ), cell constant (G^*), and observed resistance (R) of a conductivity cell is given by: (a) $\kappa = \frac{G^*}{R}$ (b) $\kappa = G^* \times R$ (c) $\kappa = \frac{R}{G^*}$ (d) $\kappa = G^* + R$ * **Chapter:** Electrochemistry | **Confidence:** High

Q3. For a first-order reaction, a plot of $\ln[R]$ against time (t) yields a straight line. The slope of this line is equal to: (a) k (b) $-k$ (c) $\frac{k}{2.303}$ (d) $\frac{-k}{2.303}$ * **Chapter:** Chemical Kinetics | **Confidence:** Very High

Q4. Which of the following elements is a transition metal but does NOT exhibit variable oxidation states? (a) Iron (Fe) (b) Scandium (Sc) (c) Copper (Cu) (d) Manganese (Mn) * **Chapter:** d- and f-Block Elements | **Confidence:** High

Q5. What is the oxidation state of the central Cobalt atom in the complex $[Co(NH_3)_5Cl]Cl_2$? (a) +2 (b) +3 (c) +4 (d) +1 * **Chapter:** Coordination Compounds | **Confidence:** Very High

Q6. Which of the following alkyl halides will undergo hydrolysis via the S_N1 mechanism at the fastest rate? (a) $CH_3 - CH_2 - CH_2 - Cl$ (b) $CH_3 - CH(Cl) - CH_3$ (c) $(CH_3)_3C - Cl$ (d) $CH_3 - Cl$ * **Chapter:** Haloalkanes and Haloarenes | **Confidence:** Very High

Q7. To synthesize tert-butyl methyl ether using Williamson's synthesis, the correct choice of reagents is: (a) Sodium methoxide and tert-butyl chloride (b) Sodium tert-butoxide and methyl chloride (c) Methanol and tert-butyl alcohol with H_2SO_4 (d) Sodium methoxide and isobutylene * **Chapter:** Alcohols, Phenols and Ethers | **Confidence:** Very High

Q8. Which of the following compounds gives a positive Fehling's test but a negative Iodoform test? (a) Ethanal (b) Propanone (c) Propanal (d) Acetophenone * **Chapter:** Aldehydes, Ketones and Carboxylic Acids | **Confidence:** High

Q9. The correct decreasing order of basic strength of the following ethylamines in an **aqueous solution** is: (a) $(C_2H_5)_2NH > (C_2H_5)_3N > C_2H_5NH_2 > NH_3$ (b) $(C_2H_5)_3N > (C_2H_5)_2NH > C_2H_5NH_2 > NH_3$ (c)

$C_2H_5NH_2 > (C_2H_5)_2NH > (C_2H_5)_3N > NH_3$ (d) $(C_2H_5)_2NH > C_2H_5NH_2 > NH_3 > (C_2H_5)_3N$ * **Chapter:** Amines | **Confidence:** Very High

Q10. The two monosaccharide units in a sucrose molecule are joined together by which specific linkage? (a) $\beta - 1, 4$-glycosidic linkage (b) $\alpha - 1, 4$-glycosidic linkage (c) $\alpha, \beta - 1, 2$-glycosidic linkage (d) Phosphodiester linkage * **Chapter:** Biomolecules | **Confidence:** High

Q11. The best reagent to convert a primary alcohol into an alkyl chloride, yielding escapable gaseous by-products, is: (a) PCl_5 (b) PCl_3 (c) Anhydrous $ZnCl_2 + HCl$ (d) $SOCl_2$ in the presence of pyridine * **Chapter:** Haloalkanes and Haloarenes | **Confidence:** Very High

Q12. Which of the following ligands causes the maximum crystal field splitting (Δ_o) in an octahedral complex? (a) CO (b) NH_3 (c) F^- (d) H_2O * **Chapter:** Coordination Compounds | **Confidence:** Very High

Directions for Q13 to Q16: Select the correct answer from the codes (a), (b), (c), and (d). *(a) Both A and R are true and R is the correct explanation of A. (b) Both A and R are true but R is NOT the correct explanation of A. (c) A is true but R is false. (d) A is false but R is true.*

Q13. Assertion (A): The rate of a chemical reaction generally increases with an increase in temperature. **Reason (R):** The activation energy of the reaction strictly decreases with an increase in temperature. * **Chapter:** Chemical Kinetics | **Confidence:** High

Q14. Assertion (A): Vitamins B and C must be supplied regularly in our diet. **Reason (R):** Vitamins B and C are water-soluble and are readily excreted in urine, hence they cannot be stored in the body. * **Chapter:** Biomolecules | **Confidence:** Very High

Q15. Assertion (A): Aniline does not undergo the Friedel-Crafts alkylation reaction. **Reason (R):** Aniline acts as a Lewis base and forms a stable salt with the Lewis acid catalyst ($AlCl_3$), deactivating the benzene ring. * **Chapter:** Amines | **Confidence:** Very High

Q16. Assertion (A): The addition of sodium chloride to water lowers its freezing point. **Reason (R):** The vapour pressure of a solution containing a non-volatile solute is higher than that of the pure solvent. * **Chapter:** Solutions | **Confidence:** High

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SECTION B: Very Short Answer Questions (2 Marks Each x 5 = 10 Marks)

Q17. Explain why a mixture of ethanol and acetone shows a positive deviation from Raoult's law. What happens to the temperature of the mixture upon mixing? * **Chapter:** Solutions | **Confidence:** Very High

Q18. Write the systematic IUPAC name and predict the hybridization of the central metal ion in the complex $[Co(C_2O_4)_3]^{3-}$. (Given: Atomic number of Co = 27). * **Chapter:** Coordination Compounds | **Confidence:** High

Q19. For an elementary reaction $A + 2B \rightarrow C$, the rate law is $Rate = k[A][B]^2$. How will the rate of the reaction be affected if the concentration of A is doubled and the concentration of B is halved simultaneously? * **Chapter:** Chemical Kinetics | **Confidence:** Very High

Q20. Out of 1-bromobutane and 2-bromobutane, identify the chiral molecule and justify your answer by defining chirality. * **Chapter:** Haloalkanes and Haloarenes | **Confidence:** High

Q21. Transition elements are well known for forming a large number of complex coordination compounds. State any two reasons to account for this observation. * **Chapter:** d- and f-Block Elements | **Confidence:** High

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SECTION C: Short Answer Questions (3 Marks Each x 7 = 21 Marks)

Q22. Differentiate between DNA and RNA on the basis of: (i) The pentose sugar present. (ii) The nitrogenous bases present. (iii) Their overall 3D macroscopic structure. * **Chapter:** Biomolecules | **Confidence:** Very High

Q23. Describe the step-by-step mechanism for the acid-catalysed dehydration of ethanol to yield ethoxyethane (diethyl ether) at 413 K. * **Chapter:** Alcohols, Phenols and Ethers | **Confidence:** Very High

Q24. Outline the chemical equations for the following specific organic conversions: (i) Propanone to Propene (ii) Benzoic acid to Benzaldehyde (iii) Ethanol to 3-Hydroxybutanal * **Chapter:** Aldehydes, Ketones and Carboxylic Acids | **Confidence:** Very High

Q25. The limiting molar conductivities (Λ_m°) for $NaCl$, HCl, and CH_3COONa are 126.4, 425.9, and 91.0 $S\,cm^2\,mol^{-1}$ respectively. Calculate the limiting molar conductivity of acetic acid (CH_3COOH). State the law that helps in making this calculation. * **Chapter:** Electrochemistry | **Confidence:** Very High

Q26. Give one simple and distinctly observable chemical test to distinguish between the following pairs of amines: (i) Ethylamine and Aniline (ii) Aniline and Benzylamine (iii) N-methylethanamine (secondary) and N,N-dimethylethanamine (tertiary) * **Chapter:** Amines | **Confidence:** High

Q27. Account for the following observations regarding d- and f-block elements: (i) The E° value for the Mn^{3+}/Mn^{2+} couple is much more positive than that for the Cr^{3+}/Cr^{2+} couple. (ii) The highest oxidation states of transition metals are exclusively shown in their oxides and fluorides. (iii) The decrease in atomic size from element to element is more pronounced in actinoids (actinoid contraction) than in lanthanoids. * **Chapter:** d- and f-Block Elements | **Confidence:** Very High

Q28. Calculate the molar mass of a polymer if 1.0 g of the polymer dissolved in 200 mL of water produces an osmotic pressure of $2.0 \times 10^{-3}\,atm$ at 300 K. (Given: $R = 0.0821\,L\,atm\,K^{-1}\,mol^{-1}$). * **Chapter:** Solutions | **Confidence:** High

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SECTION D: Case Study-Based Questions (4 Marks Each x 2 = 8 Marks)

Q29. Case Study 1: The Electrochemical Theory of Rusting Corrosion is essentially an electrochemical phenomenon. In the rusting of iron, the metal surface acts as a miniature galvanic cell in the presence of water containing dissolved oxygen and carbon dioxide. At a particular spot on the iron object, oxidation takes place, and that spot behaves as the anode. The released electrons move through the metal to another spot where they reduce oxygen in the presence of H^+ ions (which act as a catalyst), forming the cathode. The overall process leads to the formation of hydrated ferric oxide ($Fe_2O_3 \cdot xH_2O$), commonly known as rust. **(i)** Write the specific half-cell oxidation reaction that takes place at the anode during the rusting of iron. (1 Mark) **(ii)** Write the specific half-cell reduction reaction that takes place at the cathode during the rusting of iron. (1 Mark) **(iii)** How does the presence of an electrolyte, such as $NaCl$ in seawater, affect the rate of rusting? Explain the concept of 'Galvanization' as a method to prevent rusting. (2 Marks) * **Chapter:** Electrochemistry | **Confidence:** Very High

Q30. Case Study 2: Elimination vs Substitution in Haloalkanes When a haloalkane with a β -hydrogen atom is heated with an alcoholic solution of potassium hydroxide, elimination of a hydrogen halide takes place resulting in the formation of an alkene. This is known as a β-elimination reaction. If there is a possibility of forming more than one alkene, the major product is determined by a specific empirical rule. However, if the same haloalkane is treated with an *aqueous* solution of potassium hydroxide, a substitution reaction predominates, yielding an alcohol. The competition between substitution and elimination depends heavily on the nature of the base and the solvent used. **(i)** Name the specific empirical rule that predicts the major alkene formed during the dehydrohalogenation of an unsymmetrical haloalkane. (1 Mark) **(ii)** Identify the major alkene formed when 2-bromopentane is heated with alcoholic KOH. (1 Mark) **(iii)** Explain structurally and electronically why treating an alkyl halide with *aqueous* KOH leads to substitution (forming an alcohol), whereas *alcoholic* KOH leads to elimination (forming an alkene). (2 Marks) * **Chapter:** Haloalkanes and Haloarenes | **Confidence:** Very High

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SECTION E: Long Answer Questions (5 Marks Each x 3 = 15 Marks)

Q31. (A) Chemical Kinetics: Derivations & Rate Constants (i) Derive the integrated rate equation for a first-order reaction: $k = \frac{2.303}{t} \log \frac{[R]_0}{[R]}$. (2 Marks) (ii) Show mathematically that the half-life ($t_{1/2}$) of a first-order reaction is independent of the initial concentration of the reactant. (1 Mark) (iii) A first-order reaction takes 40 minutes for 30% decomposition. Calculate its half-life period. (Given: $\log 10 = 1$, $\log 7 = 0.845$). (2 Marks) * **OR (B)** (i) Define the terms 'Order of a reaction' and 'Molecularity of a reaction'. Give two points of difference between them. (2 Marks) (ii) The rate of a reaction triples when the temperature is increased from 300 K to 320 K. Calculate the activation energy (E_a) for the reaction. (Given: $R = 8.314\, J\, K^{-1}\, mol^{-1}$, $\log 3 = 0.4771$). (3 Marks) * **Chapter:** Chemical Kinetics | **Confidence:** Very High

Q32. (A) d- and f-Block Elements: Redox Reactions & Properties (i) Why is Cerium(IV) (Ce^{4+}) widely used as an excellent oxidizing agent in analytical chemistry, whereas Europium(II) (

Eu^{2+}) acts as a strong reducing agent? (2 Marks) (ii) Potassium permanganate ($KMnO_4$) acts as a powerful oxidizing agent in acidic mediums. Write the balanced complete ionic equations for the oxidation of: a) Iodide ions (I^-) to Iodine (I_2) in an acidic medium. b) Oxalate ions ($C_2O_4^{2-}$) to Carbon dioxide (CO_2) in an acidic medium. (3 Marks) * **OR (B)** (i) Describe the industrial preparation of Potassium permanganate ($KMnO_4$) from pyrolusite ore (MnO_2). Write the balanced chemical equations for the two major steps involved. (3 Marks) (ii) Complete and balance the following chemical equations: a) $Cr_2O_7^{2-} + Sn^{2+} + H^+ \rightarrow$ b) $MnO_4^- + S_2O_3^{2-} + H_2O \rightarrow$ (in faintly alkaline medium) (2 Marks) * **Chapter:** d- and f-Block Elements | **Confidence:** High

Q33. (A) Aldehydes, Ketones and Carboxylic Acids: Structural Identification An aromatic organic compound **(A)** with the molecular formula C_8H_8O gives a positive 2,4-DNP test. It forms a yellow precipitate of iodoform on treatment with iodine and sodium hydroxide, yielding the sodium salt of a carboxylic acid **(B)**. Compound **(A)** does not give a silver mirror with Tollens' reagent. However, upon vigorous oxidation with potassium permanganate ($KMnO_4/KOH$) followed by acidification, compound **(A)** yields a white crystalline carboxylic acid **(C)** with the formula $C_7H_6O_2$. (i) Deduce the structures of compounds **(A)**, **(B)**, and **(C)** and write their IUPAC names. (3 Marks) (ii) Write the balanced chemical equation for the iodoform reaction of compound **(A)** to form compound **(B)**. (1 Mark) (iii) How will you convert compound **(C)** to benzene? Write the reaction. (1 Mark) * **OR (B)** Provide the balanced chemical equations to illustrate the following important name reactions: (i) Wolff-Kishner reduction (ii) Aldol condensation (showing the formation of the α, β-unsaturated product using ethanal) (iii) Cannizzaro reaction (using formaldehyde) (iv) Rosenmund reduction (v) Etard reaction * **Chapter:** Aldehydes, Ketones and Carboxylic Acids | **Confidence:** Very High

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CBSE Class 12 (2027) — AI-Predicted via PYQ/Blueprint Analysis. Probability-weighted guidance, not actual exam questions.